Operator Discovery Report: det_double_bridge_then_segment_reversal_escape

- Operator type: perturbation.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic stagnation escape: apply a strong double-bridge perturbation first, then escalate to a structured segment reversal and occasional window shuffle, tuned to help escape two-cluster bottlenecks and nearest-nei
- Novelty classification: stagnation_escalator.
- Rediscovery signature: perturbation:double_bridge:segment_reversal:4:3.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Track stagnation count during local search and restarts.
- Use double_bridge by default and escalate toward segment_reversal after the stagnation threshold.
- Increase perturbation strength deterministically when escape pressure rises.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 976fef93ad14cb16.

Why It Should Help

- The operator targets local-minimum escape directly by controlling when and how restarts are perturbed.

Failure Cases

- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 56.2351, distance-eval delta 0.0.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 365.134986, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.0, runtime delta -37.980257, distance-eval delta -618.857143.
- sparse_three_opt gap delta 0.0, runtime delta -0.182471, distance-eval delta 0.0.
- clustered_local_search gap delta 0.015362, runtime delta -22.454986, distance-eval delta -1715.428571.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.0.
- Runtime inflation: -0.005965.
- Distance-evaluation inflation: 0.0.
- Same-gap-faster flag: True.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.0.
- simplified_perturbation mean gap 0.121317 and delta vs full operator 0.0.
- alternate_perturbation mean gap 0.121317 and delta vs full operator 0.0.